

Forensic Investigation Report

Case ID	CASE_2CF24DBA5F	Report ID	RPT-2CF24DBA5F-EE5F9A2D
Generated (UTC)	2026-08-02T14:26:08.577Z	Report version	v1

Every statement in this report references stored forensic findings; no content is generated outside computed results. Section "Report Provenance & Integrity" lists the SHA-256 digests of the source artifacts.

Executive Summary

- Case CASE_2CF24DBA5F contains 4 evidence item(s), each acquired under SHA-256 chain-of-custody verification.
- The weighted correlation engine found 6 related evidence pair(s) out of 6 analysed [correlation_analysis.json].
- Observed attack progression: financial_transaction -> post_attack -> initial_contact -> social_engineering -> credential_theft [timeline_analysis.json].

Scope & Methodology

- **objective:** Acquire, verify, correlate and reconstruct the digital evidence for this case, and derive investigative leads (suspect anchors, campaigns, cross-case links) strictly from stored, hash-verified artifacts.
- **evidence scope:** 4 evidence item(s) acquired through the CIIS intake pipeline under SHA-256 chain-of-custody control.
- **methodology:**
 - Phase 1 - Acquisition & OCR: PaddleOCR PP-OCRV5 text extraction with per-item confidence scoring; SHA-256 fingerprint recorded at intake and re-verified at read.
 - Phase 1 - Forensics: metadata/EXIF consistency, forgery signals, logo detection and evidence-confidence scoring stored per item under storage/forensics/.
 - Phase 2 - Correlation: weighted entity-overlap engine scoring every evidence pair; results in correlation_analysis.json.
 - Phase 2 - Cross-case: shared-entity matching against every other analysed case (cross_case_correlation.json).
 - Phase 2 - Campaigns / Suspects / Timeline: clustering, anchor derivation and event reconstruction over the correlated evidence set.
 - Threat intelligence: URL/domain indicators scored by the configured provider (static indicator file, or the trained phishing classifier when CIIS_ML_THREAT_INTEL=1); per-indicator results in 'Model Prediction Results'.
 - Reporting: this document is assembled exclusively from the stored outputs above; it contains no free-text generation.
- **reproducibility:** Re-running the analysis against the same stored evidence reproduces every figure herein; artifacts are versioned and never overwritten.

Case Overview

- **case id:** CASE_2CF24DBA5F
- **evidence count:** 4
- **first evidence:** 2026-07-25T12:50:00.695Z
- **last evidence:** 2026-07-25T12:53:47.011Z
- **file types:**
 - jpg
 - png
 - txt

Evidence Summary

Evidence	File	Acquired	OCR conf.	Integrity
EVID_00001	yeti.jpg	2026-07-25T12:50:00	100%	VERIFIED
EVID_00002	whatsapp_chat_export.txt	2026-07-25T12:53:07	100%	VERIFIED

Evidence	File	Acquired	OCR conf.	Integrity
EVID_00003	romanchat.jpg	2026-07-25T12:53:23	99%	VERIFIED
EVID_00004	phishing_email_screenshot.png	2026-07-25T12:53:47	90%	VERIFIED

SHA-256 digests for each item are listed in the Appendix (chain of custody).

Correlation Analysis

- **pair count:** 6
- **related pair count:** 6
- **strength distribution:**
 - **WEAK:** 6
- **top relationships:**
 - **pair:** EVID_00001 <-> EVID_00002
 - **strength:** WEAK
 - **confidence:** 0.2431
 - **explanation:** EVID_00001 (yeti.jpg) and EVID_00002 (whatsapp_chat_export.txt) show a weak relationship (confidence 0.24) based on 2 independent factor(s). Both items reference the same money: npr 2000 [weight 0.05]. Acquired 0.1 hours apart (within the 48h proximity window) [weight 0.40].
 - **pair:** EVID_00001 <-> EVID_00003
 - **strength:** WEAK
 - **confidence:** 0.2212
 - **explanation:** EVID_00001 (yeti.jpg) and EVID_00003 (romanchat.jpg) show a weak relationship (confidence 0.22) based on 1 independent factor(s). Acquired 0.1 hours apart (within the 48h proximity window) [weight 0.40].
 - **pair:** EVID_00001 <-> EVID_00004
 - **strength:** WEAK
 - **confidence:** 0.2212
 - **explanation:** EVID_00001 (yeti.jpg) and EVID_00004 (phishing_email_screenshot.png) show a weak relationship (confidence 0.22) based on 1 independent factor(s). Acquired 0.1 hours apart (within the 48h proximity window) [weight 0.40].
 - **pair:** EVID_00002 <-> EVID_00003
 - **strength:** WEAK
 - **confidence:** 0.2212
 - **explanation:** EVID_00002 (whatsapp_chat_export.txt) and EVID_00003 (romanchat.jpg) show a weak relationship (confidence 0.22) based on 1 independent factor(s). Acquired 0.0 hours apart (within the 48h proximity window) [weight 0.40].
 - **pair:** EVID_00002 <-> EVID_00004
 - **strength:** WEAK
 - **confidence:** 0.2212
 - **explanation:** EVID_00002 (whatsapp_chat_export.txt) and EVID_00004 (phishing_email_screenshot.png) show a weak relationship (confidence 0.22) based on 1 independent factor(s). Acquired 0.0 hours apart (within the 48h proximity window) [weight 0.40].

Cross-Case Correlation

No cross-case correlations were found for this case.

Campaign Analysis

- **campaign count:** 0
- **unclustered evidence:**
 - EVID_00001
 - EVID_00002
 - EVID_00003
 - EVID_00004

- **campaigns:**
 - none

Timeline Analysis

- **summary:** 4 event(s) spanning 4851.7 hour(s); 0 timestamp(s) unresolved. Observed scam progression: financial_transaction -> post_attack -> initial_contact -> social_engineering -> credential_theft.
- **stage progression:**
 - financial_transaction
 - post_attack
 - initial_contact
 - social_engineering
 - credential_theft
- **progression consistent:** False
- **milestones:**
 - **timestamp:** 2026-01-04T09:12:00+00:00
 - **description:** Investigation start - first reconstructed evidence event
 - **timestamp:** 2026-07-25T12:53:07.182000+00:00
 - **description:** First observation of stage 'financial_transaction' (EVID_00002)
 - **timestamp:** 2026-07-25T12:53:07.182000+00:00
 - **description:** First observation of stage 'post_attack' (EVID_00002)
 - **timestamp:** 2026-07-25T12:53:23.174000+00:00
 - **description:** First observation of stage 'initial_contact' (EVID_00003)
 - **timestamp:** 2026-07-25T12:53:23.174000+00:00
 - **description:** First observation of stage 'social_engineering' (EVID_00003)
 - **timestamp:** 2026-07-25T12:53:23.174000+00:00
 - **description:** First observation of stage 'credential_theft' (EVID_00003)
- **critical events:**
 - **timestamp:** 2026-01-04T09:12:00+00:00
 - **evidence id:** EVID_00001
 - **reasons:**
 - contains money entity/entities: NPR 2000
 - **timestamp:** 2026-07-25T12:53:07.182000+00:00
 - **evidence id:** EVID_00002
 - **reasons:**
 - contains money entity/entities: NPR 2000, NPR 500000

Suspect Assessment

No suspect anchors were derived from the evidence.

Threat Intelligence Summary

- **intel available:** 1.0
- **indicators checked:** 0.0
- **malicious indicators:** 0.0
- **suspicious indicators:** 0.0
- **benign indicators:** 0.0
- **evidence with threats:** 0.0
- **threat evidence ratio:** 0.0

Model Prediction Results

The active threat-intelligence provider returned no classification for any URL/domain indicator in this case's evidence.

Evidence Quality Summary

- mean image quality: 0.0
- mean evidence confidence: 78.75
- mean forgery score: 0.0
- max forgery score: 0.0
- mean ocr confidence: 0.97
- hash verified count: 4.0

Metadata Summary

- evidence id: EVID_00001
- has exif: False
- device:
- software:
- consistency notes:
 - Image carries no EXIF metadata (typical for screenshots and messaging-app exports; one less authenticity signal).
- evidence id: EVID_00002
- has exif: False
- device:
- software:
- consistency notes:
 - none
- evidence id: EVID_00003
- has exif: False
- device:
- software:
- consistency notes:
 - Image carries no EXIF metadata (typical for screenshots and messaging-app exports; one less authenticity signal).
- evidence id: EVID_00004
- has exif: False
- device:
- software:
- consistency notes:
 - Image carries no EXIF metadata (typical for screenshots and messaging-app exports; one less authenticity signal).

Investigation Statistics

- entity statistics:
 - dates: 1
 - money: 3
 - times: 3
- campaign statistics:
 - campaign count: 0.0
 - largest campaign size: 0.0
 - clustered evidence: 0.0
 - unclustered evidence: 4.0
 - mean campaign confidence: 0.0
- timeline statistics:
 - event count: 4.0
 - resolved event count: 4.0
 - unresolved event count: 0.0
 - inferred event count: 3.0
 - stage count: 5.0
 - critical event count: 2.0
 - timeline span hours: 4851.7
- correlation statistics:

- **pair count:** 6.0
- **related pair count:** 6.0
- **mean confidence:** 0.2248
- **max confidence:** 0.2431

Confidence Analysis

- **evidence id:** EVID_00001
- **score:** 77.5
- **level:** HIGH
- **explanation:** Evidence confidence is 77.5/100 (HIGH), derived from 3 verified dimension(s). Strongest signal: hash_verification (100/100). Weakest signal: processing (30/100).
- **evidence id:** EVID_00002
- **score:** 82.5
- **level:** VERY_HIGH
- **explanation:** Evidence confidence is 82.5/100 (VERY_HIGH), derived from 3 verified dimension(s). Strongest signal: hash_verification (100/100). Weakest signal: processing (30/100).
- **evidence id:** EVID_00003
- **score:** 77.5
- **level:** HIGH
- **explanation:** Evidence confidence is 77.5/100 (HIGH), derived from 3 verified dimension(s). Strongest signal: hash_verification (100/100). Weakest signal: processing (30/100).
- **evidence id:** EVID_00004
- **score:** 77.5
- **level:** HIGH
- **explanation:** Evidence confidence is 77.5/100 (HIGH), derived from 3 verified dimension(s). Strongest signal: hash_verification (100/100). Weakest signal: processing (30/100).

Investigation Conclusion

- 4/4 evidence item(s) passed SHA-256 chain-of-custody verification.
- The evidence set is internally connected (6 weighted relationship(s)), consistent with related activity rather than isolated incidents.
- Observed stage order deviates from the canonical scam sequence; evidence acquisition order should be reviewed.

Recommendations

- Go through the 2 key moment(s) - when money moved and codes were shared - with the victim, and record what they lost.
- Low urgency - handle after the others (rated 29 out of 100).

Report Provenance & Integrity

- **report id:** RPT-2CF24DBA5F-EE5F9A2D
- **generated at:** 2026-08-02T14:26:08.577Z
- **generator:** CIIS Phase-2 reporting module (template-over-data; no free-text generation)
- **evidence set digest:** 845d2580a93c7505d88a3b735dd3a956db233d26f779d05f4a85f60abfe82c3f
- **evidence set digest note:** SHA-256 over the sorted SHA-256 digests of every evidence item; any change to the evidence set changes this value.

Source artifact digests (SHA-256):

correlation_analysis.json: 9d3d4c721739bf09a59482ed218dddb9479cae1b00b3f1a9841e6dc5efb0a5b7
 cross_case_correlation.json: 6c4b7e022465546d0555c8cb462748c5b6c81a0b9a0b5bdb505f7587bf91bc63
 campaign_analysis.json: bafd96b8a7a84730a6548bdc2d9c44e959b3f0940035566e2ce11fcd20eb2009
 suspect_assessment.json: 546313af8c6a556e6ed9edd3cbf8db06474af9d683cc1ed6eb9bc6d26a4508a8
 timeline_analysis.json: 879178d918c40f554dc69b6d0f129cad6107c1275f4f6961bd57c9668eaf3108
 analytics.json: 34d087d42294b4d4b8ae464fb80ae7c56484720e479b865312a0089c3065a5ca
 case_priority.json: f30cd476a0d3d058a919ba12443715445cad4e6a5587b85c2bf371b4bff961d8
 graph.json: 6478030850e022f70aabbe86ad1dc987508916865c7aab03093a42c99b0c0a12

Appendix

- **chain of custody:**
 - **evidence id:** EVID_00001
 - **sha256:** 8f2e60ea6fc0eeffd209e28eb29228797104b60a2505934fc3d42642c100ea79
 - **upload time:** 2026-07-25T12:50:00.695Z
 - **status:** processed
 - **evidence id:** EVID_00002
 - **sha256:** c7ab347cfbd2b80c944c85b3442f3b2c400da1d3b92962f9ded8598e1e06cfb2
 - **upload time:** 2026-07-25T12:53:07.182Z
 - **status:** processed
 - **evidence id:** EVID_00003
 - **sha256:** 0340d5863313f4cc0b9b40ed835a77a71b8304d2e08eb32f08a56ed93a6e0b10
 - **upload time:** 2026-07-25T12:53:23.174Z
 - **status:** processed
 - **evidence id:** EVID_00004
 - **sha256:** 8ef0f27a84960b656d9051892e36096ce56fe9f30eb9a935d4d4330a002b9b7f
 - **upload time:** 2026-07-25T12:53:47.011Z
 - **status:** processed
- **stored artifacts root:** storage/investigation/<CASE_ID>/
- **phase1 artifacts root:** storage/forensics/<EVIDENCE_ID>/

Prepared by (system)

Cybercrime Investigation Intelligence Engine
Automated Phase-2 reporting module

Reviewed by (investigator)

Date:

Date:

Signature: